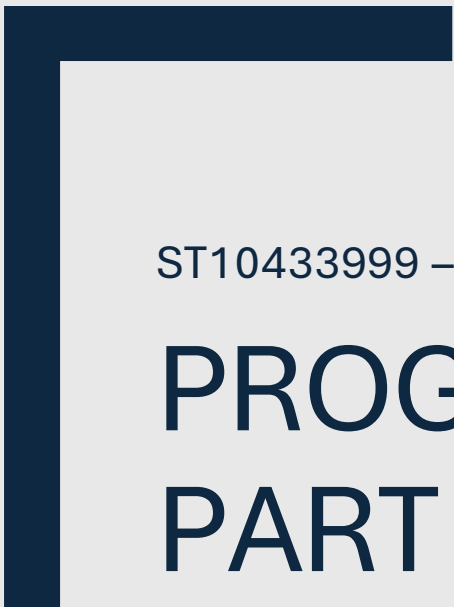
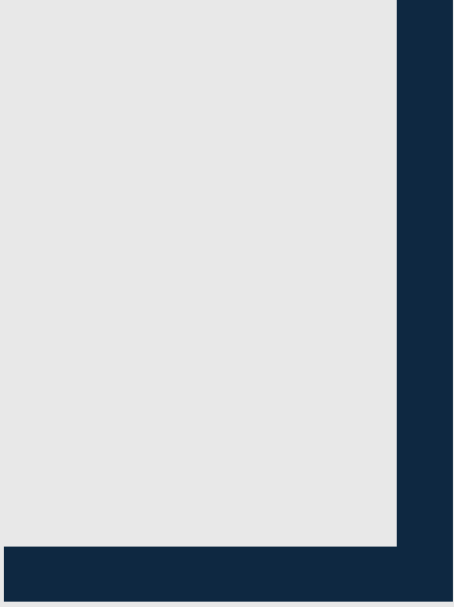




ST10433999 – Muhammed Suliman

# PROG 7312 POE 1

## PART 1



# Smart-X IoT Mesh Ecosystem – Part 1, Task 1: Research Report

## 1. Introduction

The Smart-X application will consume thousands of ESP32 nodes' telemetry data via a .NET 10 gateway. The data types would include floats for soil moisture, integers for wattage, and booleans for valve state. The individuals that would configure and debug such a system are engineers, not end-users and are just as likely to quit using a console due to its unapproachability or noise. This report enumerates five different developer-focused engagement tactics found in the literature, then justifies the approach selected for implementation in Part 2 of the portfolio: a real-time Node Pulse Board with anomaly-rank triage.

## 2. Five Engagement and Interactivity Strategies Considered

| Strategy                                                                                                   | How it makes telemetry clear and actionable                                                                                                                                                                                                                                            | Key sources                                                       |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>1. Real-time visual telemetry feedback (live sparklines and node status tiles)</b>                      | Streams the latest readings of each node into a compact chart to enable the user to observe a spike or a flat line on-the-fly instead of examining a log. Sarikaya et al. (2019) refer to this as an 'awareness' dashboard, whose value is in the immediate visual change it presents. | Shneiderman (1996); Sarikaya et al. (2019); Mofidul et al. (2022) |
| <b>2. Anomaly-highlighted alerting with severity triage (statistical thresholds instead of raw alarms)</b> | Compares each reading to a rolling baseline (z-score), prioritizes alerts by severity so that nuisance alarms don't drown out real faults - the alarm flood problem documented in industrial control rooms.                                                                            | Chandola, Banerjee and Kumar (2009); Wang et al. (2016)           |
| <b>3. Proactive alert gamification (streaks, badges and response-time targets for operators)</b>           | Turns routine acknowledgement of alerts into measurable goals. Evidence in software engineering shows gamification raises developer engagement, though it must be applied carefully to avoid trivialising critical faults.                                                             | Stol, Schaarschmidt and Goldblit (2022)                           |
| <b>4. Simplified onboarding with progressive disclosure (guided sensor registration wizard)</b>            | Reveals only the fields needed at each step (MAC → location → category → attachments), reducing the intrinsic cognitive load of configuring thousands of devices.                                                                                                                      | Sweller (1988); Stelea, Sangeorzan and Enache-David (2025)        |
| <b>5. Overview-first drill-down navigation (fleet summary → zone → node → packet)</b>                      | Applies the Visual Information-Seeking Mantra 'overview first, zoom and filter, then details-on-demand' so a technician can move from a fleet-wide health count to one packet in three clicks.                                                                                         | Shneiderman (1996); Sarikaya et al. (2019)                        |

## 3. Chosen Strategy: Real-Time Node Pulse Board with Anomaly Triage

### 3.1 Description and planning

The selected strategy incorporated strategies 1 and 2 into a single feature: a grid of live node tiles, each displaying a sparkline of the last sixty readings plus a colour-coded health state, backed by a severity-sorted triage feed (Figure 1). The state of a tile was calculated on the API from a rolling baseline: for each incoming TelemetryPacket is appended to a per-node window, the mean and standard deviation were recalculated and a z-score classified the reading as Healthy, Drift or Spike. A heartbeat timer independently marked a node Silent if no packet had arrived within three expected intervals (Figure 2). Anomalies and user-reported issues were stored in a generic List that the Blazor client polled asynchronously, so the board and the feed updated without a page refresh.

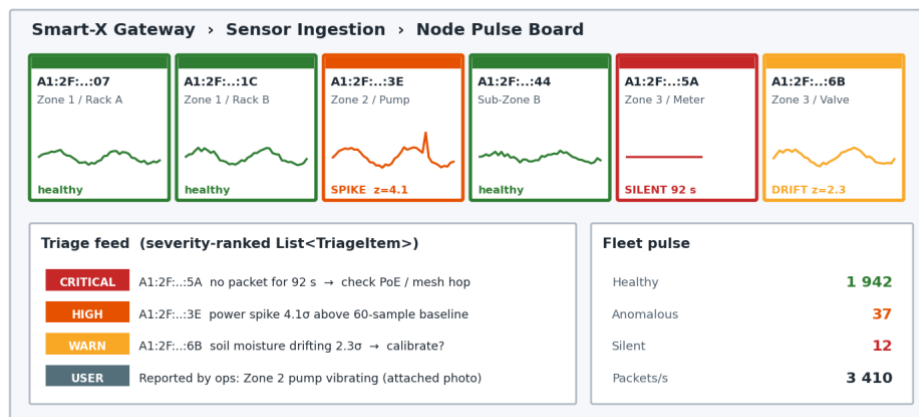


Figure 1: Concept layout of the Node Pulse Board – live tiles, severity-ranked triage feed and fleet summary.

### 3.2 Justification

The first reason is perceptual: Shneiderman's (1996) mantra of overview first, zoom and filter, and then details on demand is built right into the interface the fleet summary is the overview, the tile grid is the filtered, zoomed-in view, and clicking on a tile reveals packet-level detail. Sarikaya et al. (2019) argue that dashboards designed for operational awareness are most effective when they are able to provide an immediate visual change in response to a change in the underlying data: a tile changing from green to red satisfies this requirement in a way that a table of numbers cannot.

The second rationale focuses on anomalous spikes. Chandola, Banerjee and Kumar (2009) classify point anomalies as individual observations that differ from a learned idea of normal and demonstrate that statistical tests, including distance-based ones, are a valid approach to detect them in numeric streams. A rolling z-score is the simplest such test, as it takes  $O(1)$  operations to calculate and has no training phase, but it can still make a good detector, since it is adaptive to the distribution of each sensor. Thus, a  $4\sigma$  wattage spike on a smart meter and a  $4\sigma$  moisture increase in a hydroponic tray would both be reported as anomalies without requiring any domain-specific parameters. Mofidul et al. (2022) used a similar approach to design an IIoT energy monitoring system and found that the only way for anomalies to be actually useful was to report them alongside the normal baseline on a dashboard.

The third justification is sensor disconnections. A silent node produces no data, meaning that it cannot be detected on a chart, but the heartbeat timer exposes such nodes by translating the absence of information into a visual state. It is critical to Smart-X's operations, considering that the program utilizes thousands of nodes, making it impossible for a single technician to observe all nodes' status in one chart. Instead, the SILENT 92 s tile located at the top of the triage feed draws attention to the disconnected node, its approximate location and the cause of the silent state.

The fourth rationale is troubleshooting time (Wang et al., 2016). Industrial alarm systems fail catastrophically or chatter and waste operator time on nuisance alarms before reaching actual faults. Ranking the feed by severity and suppressing repeated assertions of the same node-state within a cooldown period shortens the list and qualifies each entry with the deployment path (Sub-Zone B → Zone 1 → Facility A) resolved by the recursive validator. The engineer walks to the indicated location while the system awaits further input, and attaching a photo or configuration file to the triage item closes the loop between field and console.

Finally, the strategy is both measurable and unique. It reflects a real system state, not a progress bar, and the time-to-acknowledge each item can later feed a light gamification layer (Stol, Schaarschmidt and Goldblit, 2022), without requiring changes to the data model. As the tiles are computed from the same List and jagged-array batches required by the brief, this engages the feature in the core data structures rather than alongside them.

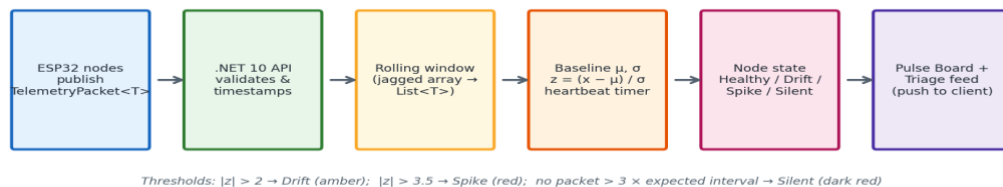


Figure 2: Anomaly and dropout detection pipeline from ESP32 packet to Pulse Board state.

## 4. Conclusion

Out of the five strategies reviewed, real-time Node Pulse Board with anomaly triage addresses the two types of failure modes mentioned in the brief data spikes and sensor disconnections – and is cheap to calculate on a resource-constrained gateway. The strategy will be implemented in the Sensor Ingestion pillar with command-stream events in Part 2.

## References

- Chandola, V., Banerjee, A. and Kumar, V. (2009) 'Anomaly detection: a survey', *ACM Computing Surveys*, 41(3), Article 15, pp. 1–58. doi:10.1145/1541880.1541882.
- Mofidul, R.B., Alam, M.M., Rahman, M.H. and Jang, Y.M. (2022) 'Real-time energy data acquisition, anomaly detection, and monitoring system: implementation of a secured, robust, and integrated global IIoT infrastructure with edge and cloud AI', *Sensors*, 22(22), Article 8980. doi:10.3390/s22228980.
- Sarikaya, A., Correll, M., Bartram, L., Tory, M. and Fisher, D. (2019) 'What do we talk about when we talk about dashboards?', *IEEE Transactions on Visualization and Computer Graphics*, 25(1), pp. 682–692. doi:10.1109/TVCG.2018.2864903.
- Shneiderman, B. (1996) 'The eyes have it: a task by data type taxonomy for information visualizations', *Proceedings of the IEEE Symposium on Visual Languages*, Boulder, CO, 3–6 September. Los Alamitos, CA: IEEE, pp. 336–343. doi:10.1109/VL.1996.545307.
- Stelea, G.A., Sangeorzan, L. and Enache-David, N. (2025) 'Accessible IoT dashboard design with AI-enhanced descriptions for visually impaired users', *Future Internet*, 17(7), Article 274. doi:10.3390/fi17070274.
- Stol, K.-J., Schaarschmidt, M. and Goldblit, S. (2022) 'Gamification in software engineering: the mediating role of developer engagement and job satisfaction', *Empirical Software Engineering*, 27(2), Article 35. doi:10.1007/s10664-021-10062-w.
- Sweller, J. (1988) 'Cognitive load during problem solving: effects on learning', *Cognitive Science*, 12(2), pp. 257–285. doi:10.1207/s15516709cog1202\_4.
- Wang, J., Yang, F., Chen, T. and Shah, S.L. (2016) 'An overview of industrial alarm systems: main causes for alarm overloading, research status, and open problems', *IEEE Transactions on Automation Science and Engineering*, 13(2), pp. 1045–1061. doi:10.1109/TASE.2015.2464234.